

Multiple Choice (10 marks)

1. A major difference between cancer cells and normal cells is that
 - (a) Cancer cells have no telomere strands
 - (b) Normal cells become cancerous when their telomere strands are used up
 - (c) Cancer cells have no upper limit on cell division
 - (d) Normal cells are protected from cancer when they have telomerase
2. All other things being equal, which of the following men is likely to be the happiest in retirement?
 - (a) Gary who is still married
 - (b) Leo who never married
 - (c) Josh who is divorced
 - (d) Caleb who is remarried
3. Which level of personality description appears to show the most change with age?
 - (a) Traits
 - (b) Personal concerns
 - (c) Coping strategies
 - (d) Identity
4. The parvoviruses have a rather unique genome; what form does it take?
 - (a) Circular ds DNA
 - (b) Small linear ss DNA
 - (c) ds DNA genome
 - (d) RNA genome with RT
5. What is the most successful and most widely used 'travel' vaccine for international visitors?
 - (a) Influenza
 - (b) MMR
 - (c) Yellow Fever
 - (d) Rabies

True / False (10 marks)

Answer True or False

6. True or False: DNA damage caused by radiation from anti-oxidants contributes to cellular senescence by altering DNA instructions.
7. True or False: The prepuce, also known as the foreskin, is the tissue that is removed during male circumcision.
8. True or False: In X-linked dominant inheritance, males can transmit the trait to both male and female offspring.
9. True or False: Adaptive immunity is characterized by a rapid response to foreign proteins, similar to innate immunity.
10. True or False: Public health ethics is primarily population-centered, focusing on the health of communities rather than individual patients.

Long Form Response (20 marks)

Answer each question with a clear and complete explanation.

11. Discuss the significance of patient positioning in healthcare, specifically focusing on the implications and benefits of having a patient lie face down during medical assessments or treatments.
12. Discuss the applications of next generation sequencing (NGS) in health sciences, analyzing its limitations and potential areas where it may not be effectively utilized.

End of Paper